

IRRIGATION HYDRAULIC PERFORMANCE DASHBOARD

SCHEME

All

BLOCKS

All

PIPE DIAMETER

All

PIPE MATERIAL

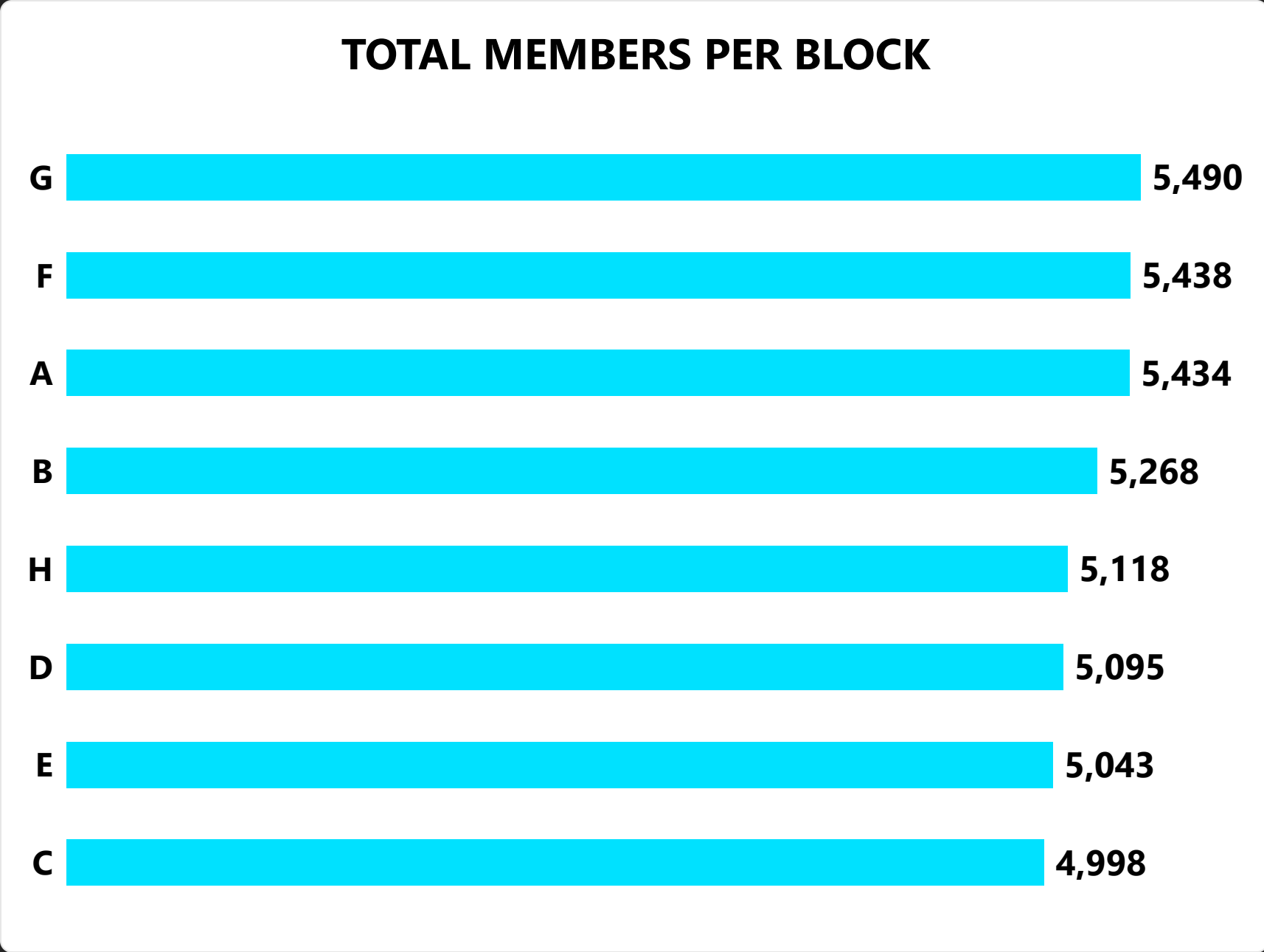
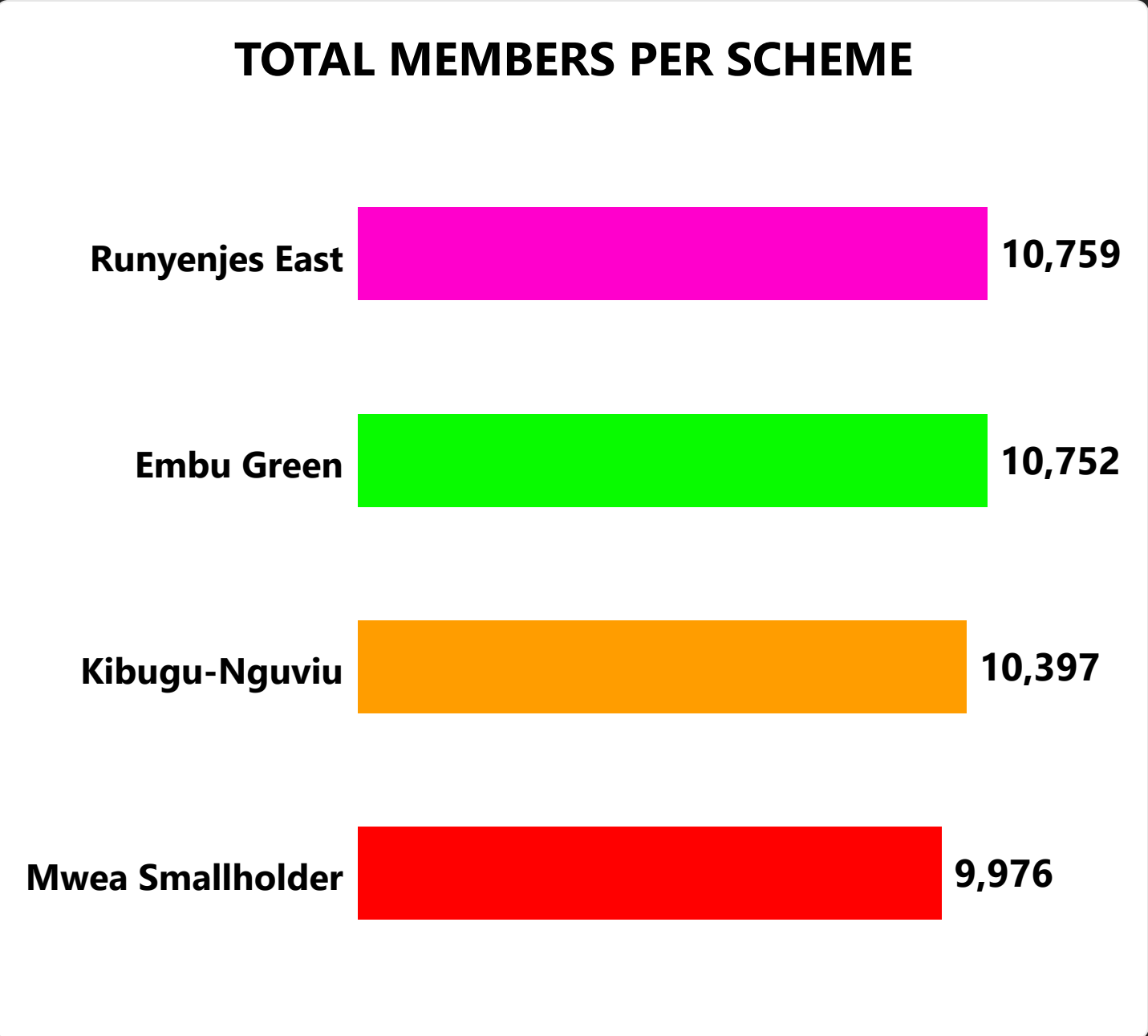
All

DATE

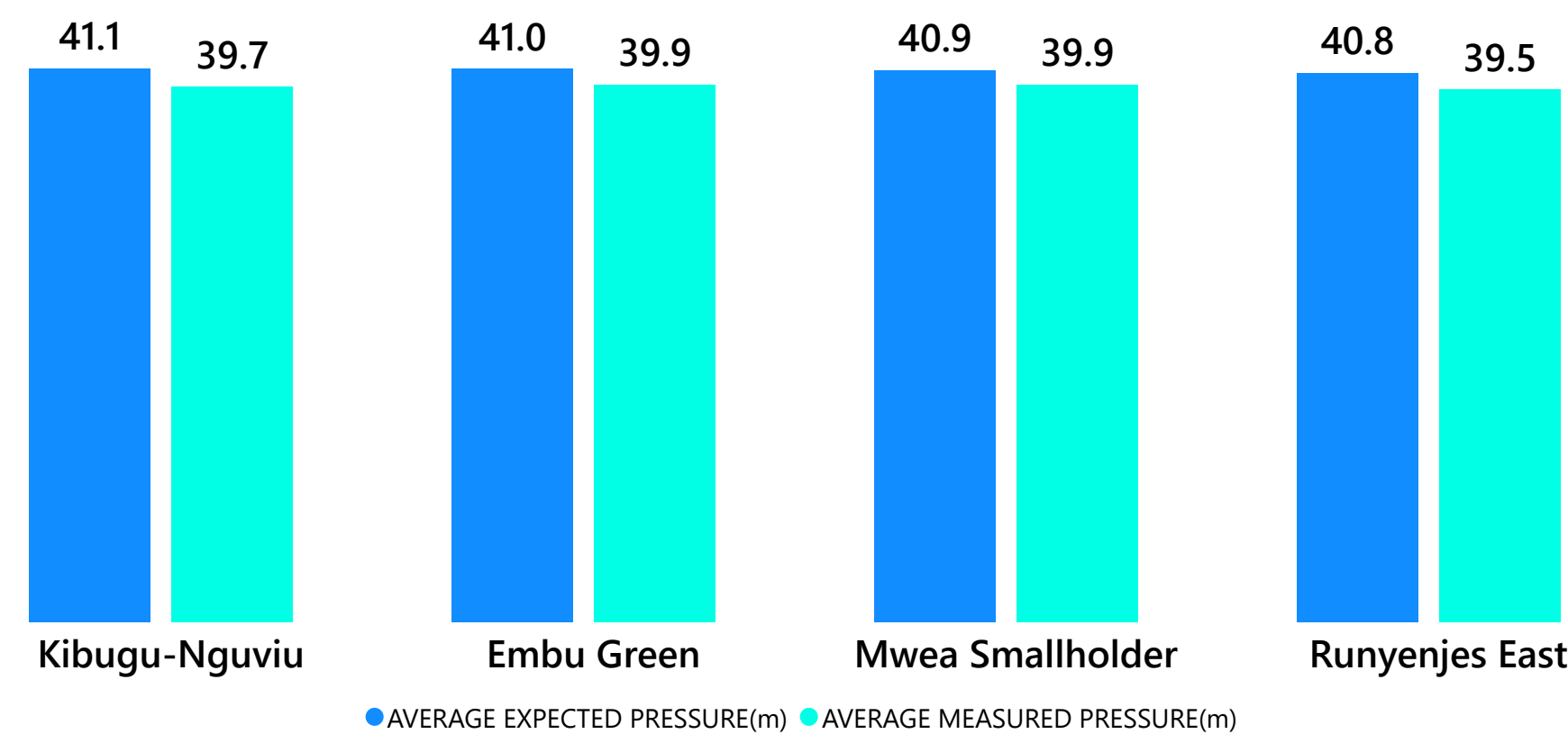
1/1/2024

8/31/2026

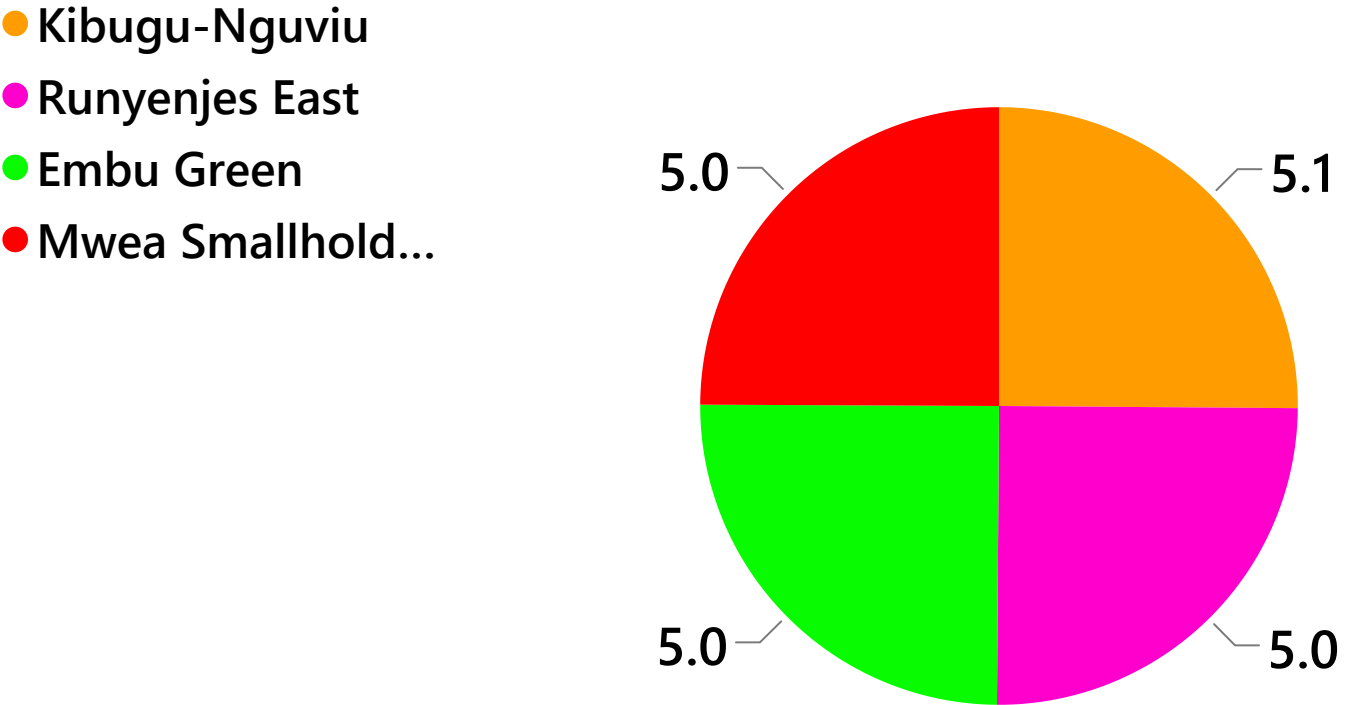
SCHEMES 4	BLOCKS 8	TOTAL MEMBERS/OFFTAKES 41,884	AVERAGE FLOW(L/s) 5.0	AVERAGE HEADLOSS(m) 2.5
AVERAGE EXPECTED PRESSURE(m) 41.0	AVERAGE MEASURED PRESSURE(m) 39.8	AVERAGE PRESSURE DEFICIT(m) -1.2	MAXIMUM MEASURED PRESSURE(m) 65.0	MINIMUM MEASURED PRESSURE(m) 11.0



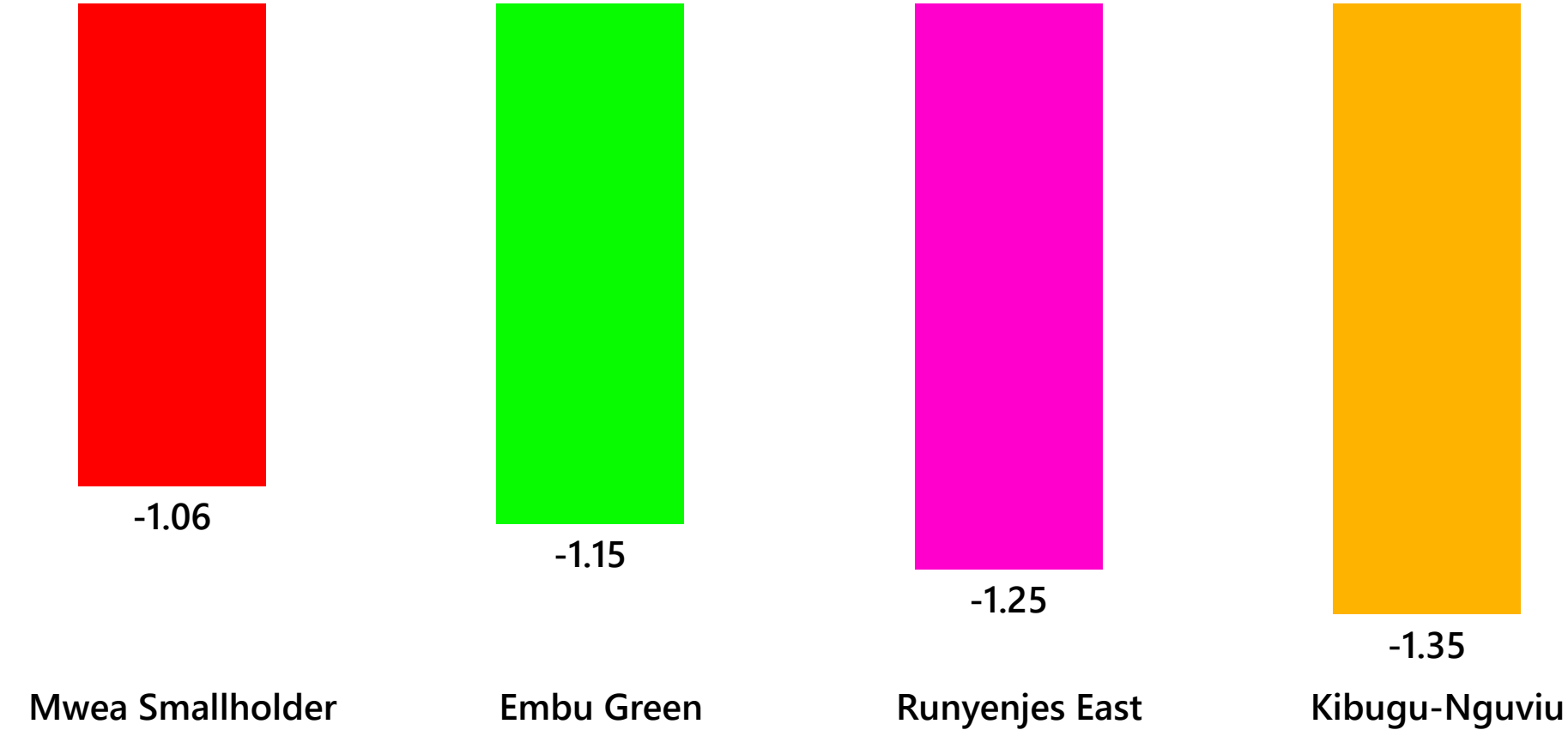
AVERAGE EXPECTED PRESSURE AND AVERAGE MEASURED PRESSURE PER SCHEME



AVERAGE FLOW PER SCHEME



AVERAGE PRESSURE DEFICIT PER SCHEME



AVERAGE HEADLOSS PER SCHEME

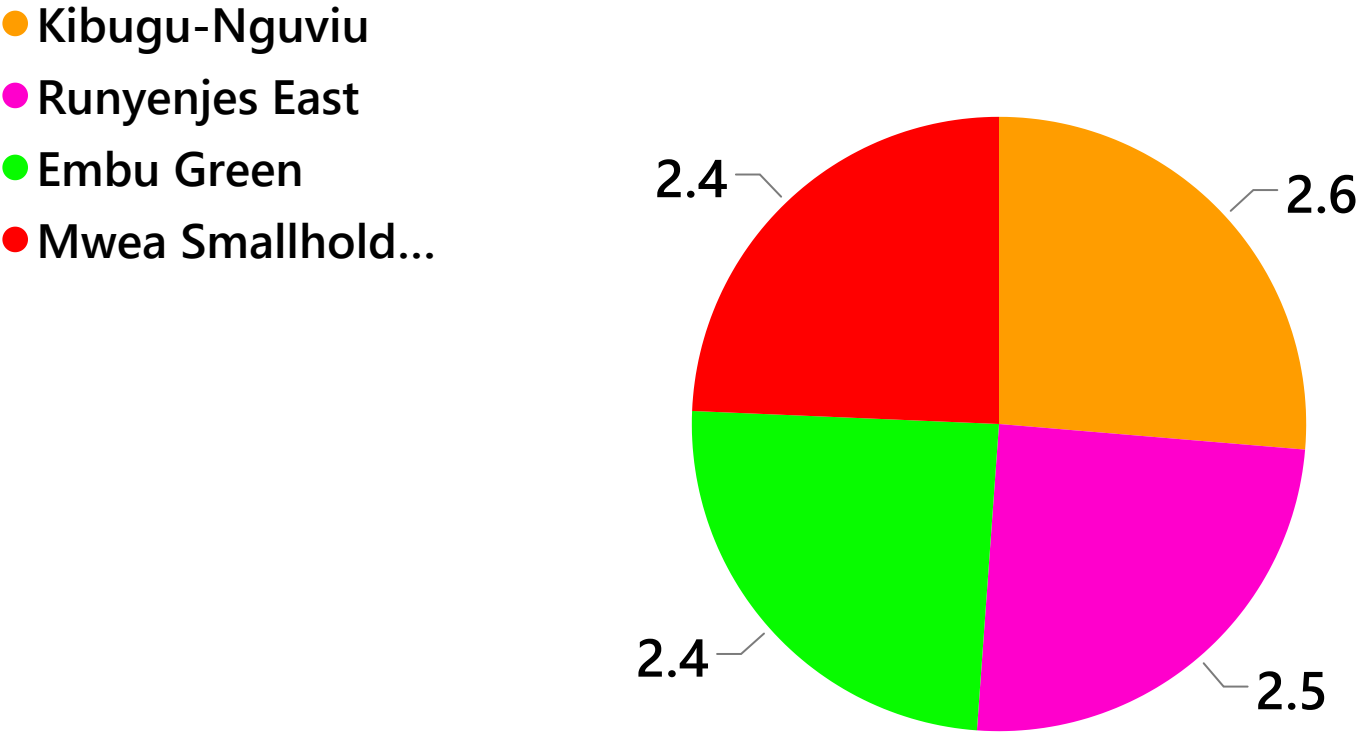


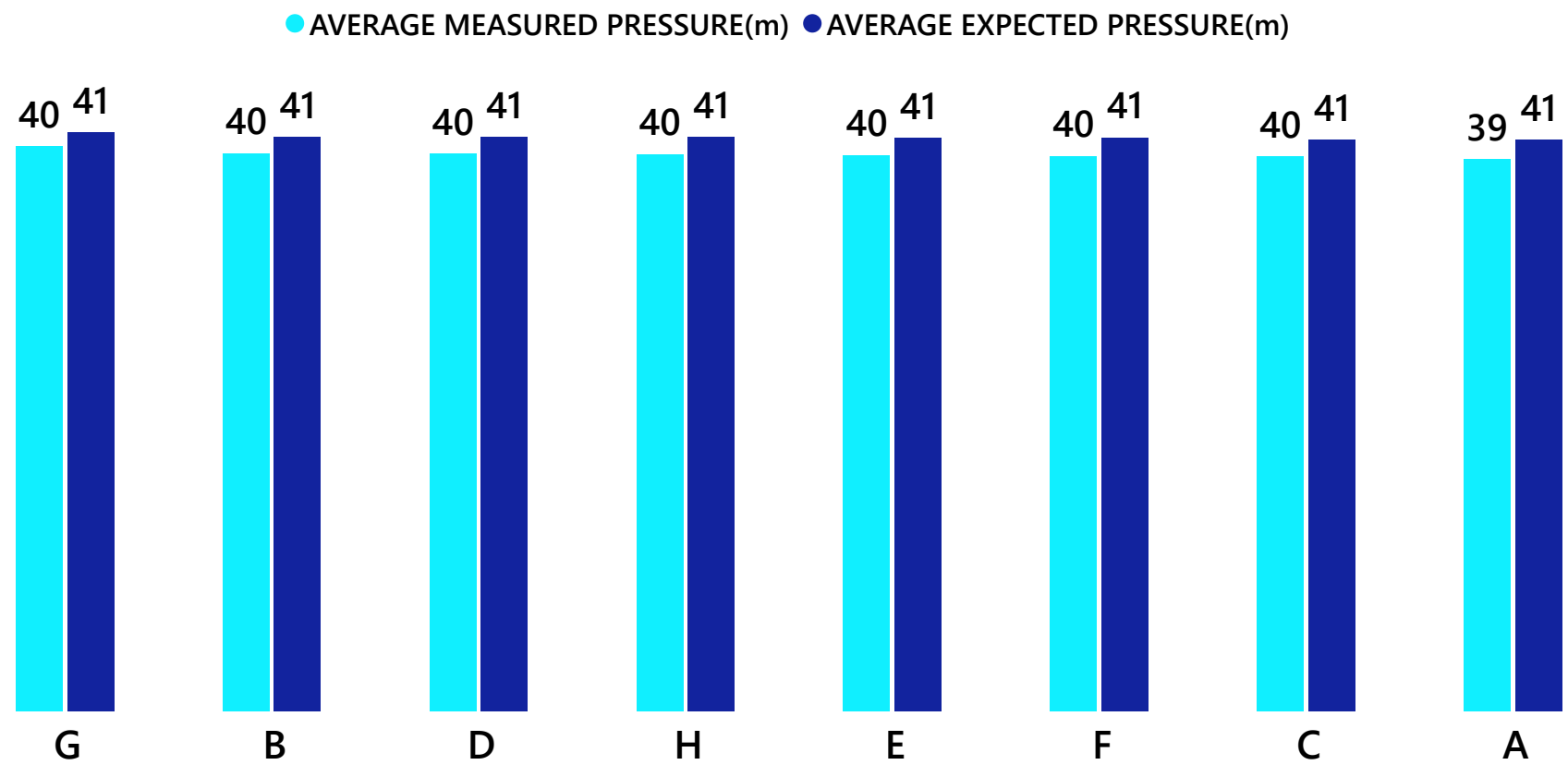
TABLE SHOWING TOP 15 CHAINAGES WITH THE HIGHEST NEGATIVE PRESSURE DEFICITS.

SCHEME	BLOCK	CHAINAGE(m)	PIPE DIAMETER(mm)	PIPE LENGTH(m)	PIPE MATERIAL	AVERAGE PRESSURE DEFICIT(m)	AVERAGE EXPECTED PRESSURE(m)	AVERAGE MEASURED PRESSURE(m)
Runyenjes East	C	4253.6	63	311.8	HDPE	-20.28	44.61	24.33
Embu Green	E	1641.3	50	436.2	PVC	-18.21	35.11	16.90
Kibugu-Nguviu	A	4344.1	32	653.8	PVC	-16.14	32.35	16.21
Mwea Smallholder	B	274.5	63	761.6	GI	-16.11	49.28	33.17
Embu Green	H	3074	32	600.9	PVC	-15.96	36.79	20.83
Kibugu-Nguviu	D	407.4	50	391.9	GI	-15.76	43.07	27.31
Kibugu-Nguviu	C	3250.1	32	184.8	HDPE	-15.55	48.48	32.93
Kibugu-Nguviu	H	3008.2	32	774.1	PVC	-15.43	26.47	11.04
Runyenjes East	G	4985.8	32	176.1	GI	-15.24	34.28	19.04
Mwea Smallholder	A	1545.1	32	490.3	PVC	-15.22	39.44	24.22
Runyenjes East	H	4218.1	63	230.7	HDPE	-15.03	49.31	34.28
Embu Green	B	2365.1	50	531.7	HDPE	-14.98	34.35	19.37
Embu Green	B	4565.6	50	329.6	PVC	-14.80	39.36	24.56
Embu Green	C	2143.1	90	425.1	PVC	-14.80	43.34	28.54
Kibugu-Nguviu	E	4708.3	75	596.5	PVC	-14.54	36.66	22.12
Embu Green	D	1772.2	75	514.4	HDPE	-14.12	35.07	20.95
Mwea Smallholder	F	3880.4	32	97.1	GI	-14.08	42.79	28.71
Embu Green	C	2711.9	63	665.3	PVC	-13.98	31.74	17.76
Runyenjes East	A	1513.2	25	788.2	PVC	-13.95	41.63	27.68
Runyenjes East	B	2459.8	50	349.5	PVC	-13.94	49.78	35.84
Runyenjes East	D	1641.3	63	184.8	GI	-13.81	48.63	34.18
Total						-1.20	40.96	39.75

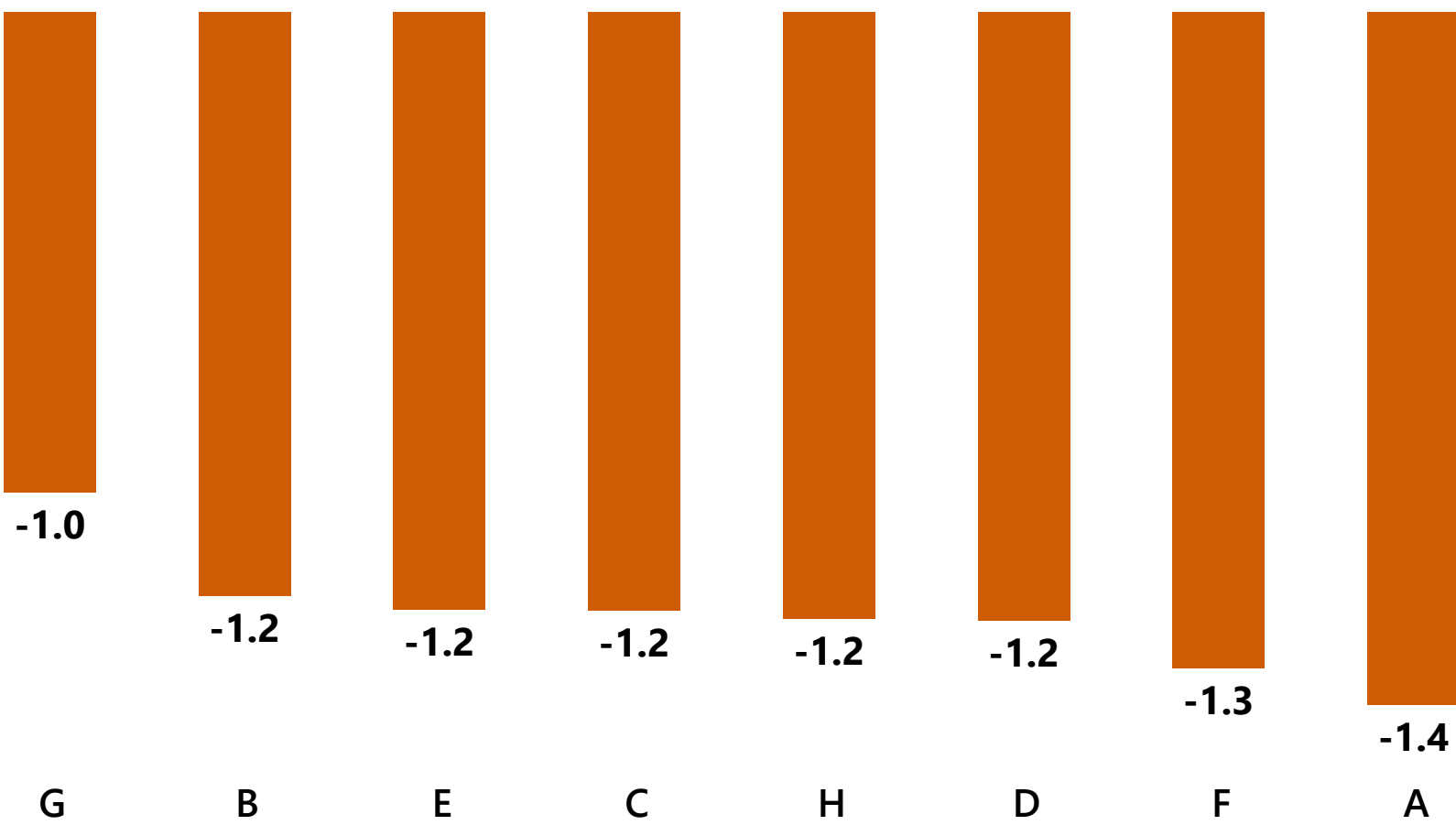
TABLE SHOWING TOP 15 CHAINAGES MARKED WITH HIGH PRESSURE HYDRAULIC STATUS.

SCHEME	BLOCK	CHAINAGE(m)	PIPE DIAMETER(mm)	PIPE MATERIAL	AVERAGE MEASURED PRESSURE(m)	AVERAGE EXPECTED PRESSURE(m)	AVERAGE PRESSURE DEFICIT(m) ▼	HYDRAULIC STATUS
Kibugu-Nguviu	G	3798.1	25	HDPE	64.32	44.34	19.98	High Pressure
Embu Green	C	2331	63	PVC	61.63	47.69	13.94	High Pressure
Embu Green	G	2256.1	75	PVC	55.87	42.40	13.47	High Pressure
Runyenjes East	F	1642.4	40	HDPE	48.73	35.40	13.33	Normal
Embu Green	D	594.8	63	HDPE	52.42	39.48	12.94	High Pressure
Mwea Smallholder	B	295.9	75	PVC	54.92	42.12	12.80	High Pressure
Embu Green	C	4603.6	50	PVC	60.89	48.39	12.50	High Pressure
Mwea Smallholder	C	2026.3	32	PVC	55.14	42.99	12.15	High Pressure
Mwea Smallholder	B	2351.4	75	PVC	50.44	38.44	12.00	High Pressure
Mwea Smallholder	E	3904.4	40	PVC	47.33	35.49	11.84	Normal
Embu Green	D	4881.7	90	PVC	44.82	33.37	11.45	Normal
Mwea Smallholder	H	2098.8	63	GI	51.49	40.12	11.37	High Pressure
Embu Green	H	902.6	90	PVC	64.16	52.80	11.36	High Pressure
Kibugu-Nguviu	D	3106	40	PVC	50.95	39.68	11.27	High Pressure
Kibugu-Nguviu	D	2735.1	63	PVC	60.47	49.25	11.22	High Pressure
Kibugu-Nguviu	E	770.9	32	PVC	51.69	40.60	11.09	High Pressure
Runyenjes East	A	1404.6	75	PVC	51.03	40.00	11.03	High Pressure
Mwea Smallholder	B	1808.4	40	PVC	50.76	39.77	10.99	High Pressure
Embu Green	H	1325.5	40	PVC	43.85	32.91	10.94	Normal
Mwea Smallholder	C	1377.5	40	PVC	57.40	46.51	10.89	High Pressure
Mwea Smallholder	H	2521	50	PVC	51.12	40.31	10.81	High Pressure
Total					39.75	40.96	-1.20	

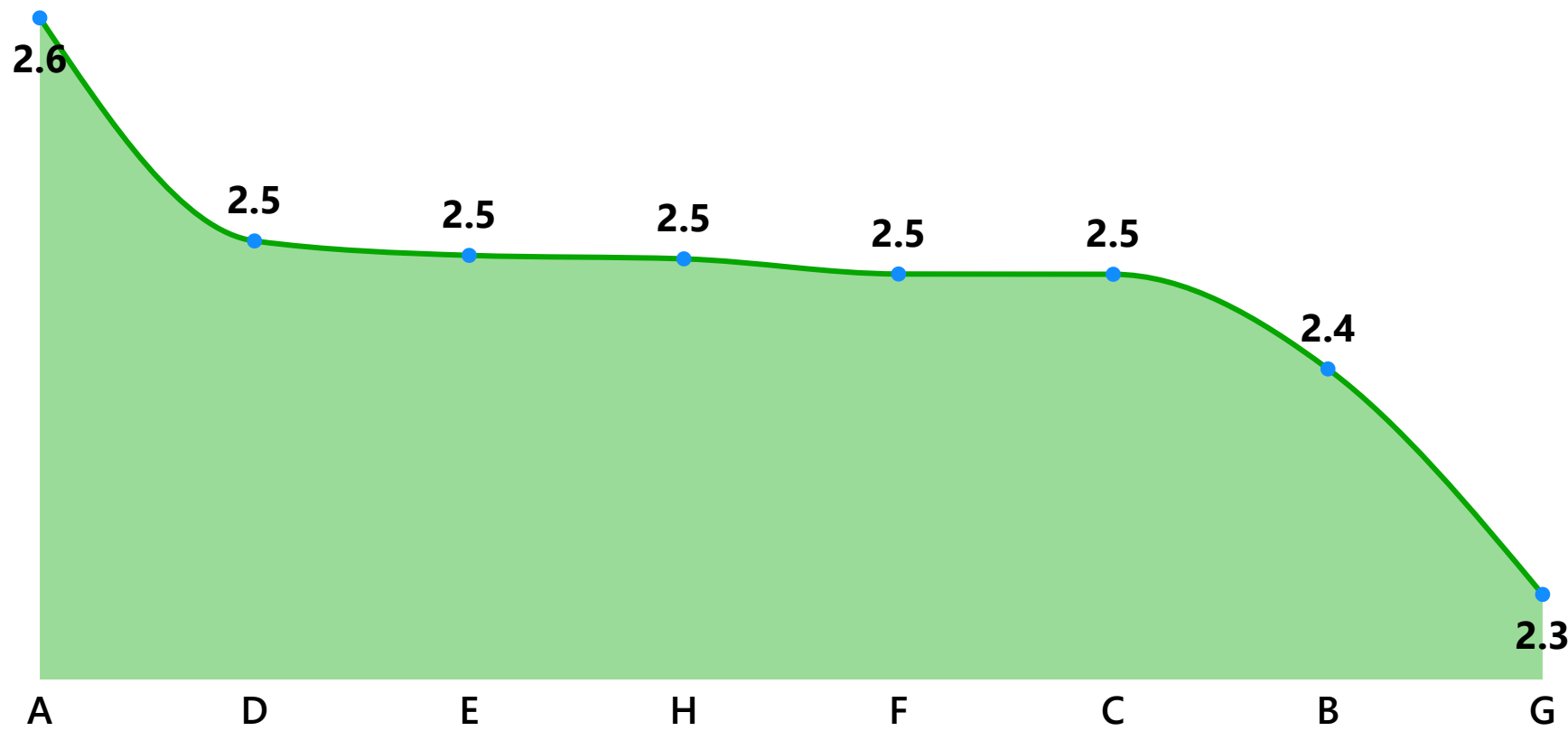
AVERAGE MEASURED PRESSURE V/s AVERAGE EXPECTED PRESSURE PER BLOCK



AVERAGE PRESSURE DEFICIT PER BLOCK



AVERAGE HEADLOSS PER BLOCK



AVERAGE FLOW(L/s) PER BLOCK

